Worksheet 2 · Prime Factors (LCM, HCF) IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Write a number as a product of its prime factors • Find the HCF of two or more numbers using prime factors • Find the LCM of two or more numbers and use it to solve problems		
Grade: / 53	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

HCF → Multiply all numbers in the middle of venn diagram

LCM → Multiply all numbers in the whole venn diagram

Questions *Prime Factors (LCM, HCF)*

Question 1 (4 marks)

(a) Find the highest common factor (HCF) of 200 and 420

(2)

$A = 2^3 \times 3 \times 5 \times 7^2$
$B = 2 \times 3^2 \times 7$
$C = 3 \times 5^2 \times 11$

(b) Find the lowest common multiple (LCM) of A, B and C
Write your answer as a product of powers of prime factors.

(2)

Question 2 (2 marks)

Find the lowest common multiple (LCM) of 28 and 105

Question 3 (4 marks)

$A = 3^2 \times 5^4 \times 7$ $B = 3^4 \times 5^3 \times 7 \times 11$

(a) Find the highest common factor (HCF) of A and B.

(2)

(b) Find the lowest common multiple (LCM) of A and B.

(2)

Question 4 (4 marks)

(a) Write 720 as a product of its prime factors.
Show your working clearly.

(3)

(b) Find the smallest whole number that 720 can be multiplied by to give a square number.

(1)

Question 5 (4 marks)

(a) Find the highest common factor (HCF) of 96 and 120

(2)

$A = 2^3 \times 5 \times 7^2 \times 11$
$B = 2^4 \times 7 \times 11$
$C = 3 \times 5^2$

(b) Find the lowest common multiple (LCM) of A, B and C.

(2)

Question 6 (3 marks)

Write 600 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 1 is 3 marks)

Question 7 (4 marks)

$A = 3^5 \times 5 \times 7^3$
$B = 2^3 \times 3 \times 7^4$

(a) (i) Find the Highest Common Factor (HCF) of A and B.

(ii) Find the Lowest Common Multiple (LCM) of A and B.

(2)

$A = 3^5 \times 5 \times 7^3$
$B = 2^3 \times 3 \times 7^4$
$C = 2^p \times 5^q \times 7^r$

Given that
the HCF of B and C is $2^3 \times 7$
the LCM of A and C is $2^4 \times 3^5 \times 5^2 \times 7^3$

(b) find the value of p, the value of q and the value of r.

$p =$ .....

$q =$ .....

$r =$ .....

(2)

Question 8 (5 marks)

$N = 480 \times 10^9$

(a) Write N as a number in standard form.

(1)

(b) Write N as a product of powers of its prime factors.
Show your working clearly.

(3)

(c) Find the largest factor of N that is an odd number.

(1)

Question 9 (4 marks)

(a) Find the highest common factor (HCF) of 56 and 84
Show your working clearly.

(2)

(b) Find the lowest common multiple (LCM) of 60 and 72
Show your working clearly.

(2)

Question 10 (5 marks)

(a) Find the Highest Common Factor (HCF) of 140 and 245

(2)

A machine has a buzzer that sounds every 50 minutes.
The machine also has a bell that sounds every 80 minutes.

The buzzer and the bell sound together at 10 am.

(b) Find the time at which they next sound together.

(3)

Question 11 (2 marks)

$A = 2^3 \times 3^2 \times 5^2 \times 11$
$B = 2^4 \times 3 \times 5^4 \times 13$

Find the lowest common multiple (LCM) of A and B.
Give your answer as a product of powers of prime numbers.

Question 12 (3 marks)

Write 3.6×10^3 as a product of powers of its prime factors.
Show your working clearly.

Question 13 (4 marks)

(a) Work out the lowest common multiple (LCM) of 36 and 120

(2)

$A = 5^2 \times 7^4 \times 11^p$
$B = 5^m \times 7^{n-5} \times 11$

m, n and p are integers such that
$m > 2$
$n > 10$
$p > 1$

(b) Find the highest common factor (HCF) of A and B
Give your answer as a product of powers of its prime factors.

(2)

Question 14 (3 marks)

Find the lowest common multiple (LCM) of 20, 30 and 45

Question 15 **(2 marks)**

$A = 23 \times 32 \times 54$

$B = 35 \times 5 \times 73$

Find the Highest Common Factor (HCF) of A and B.

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